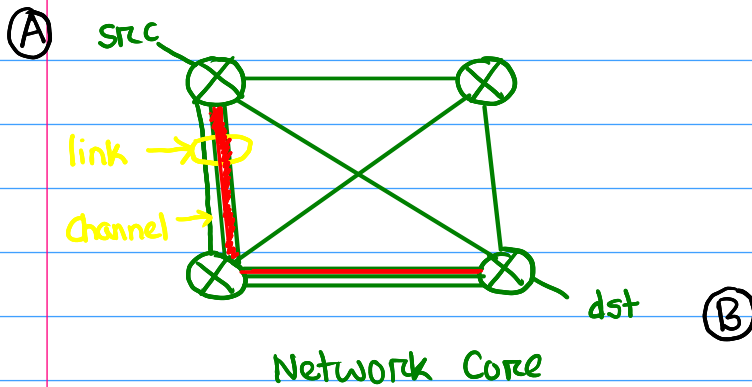


Computer Networks



- ① circuit switching → fixed + dedicated
- ① packet switching → variable

circuit switching

- FDM
- TDM

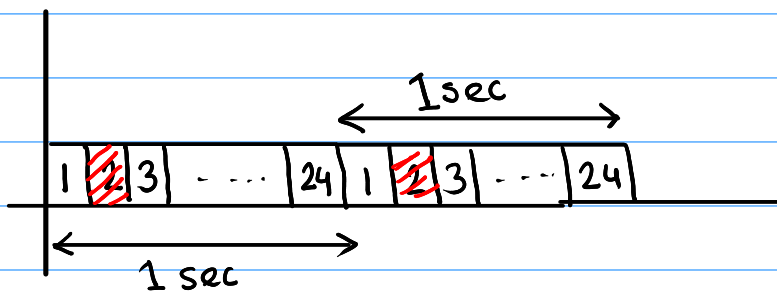
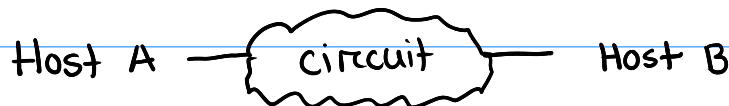
✓ **Exercise 4:** How long does it take to send a file of **640,000 bits** from host A to host B over a **circuit-switched network**?

- ❖ All links are 1.536 Mbps
- ❖ Each link uses TDM with 24 slots/sec
- ❖ 500 msec to establish end-to-end circuit

$$M = 10^6$$

b → bit
B → byte

↪ 0.5 s



$$\rightarrow 1.536 \text{ Mbps} = 1.536 \times 10^6 \text{ bps}$$

$$\text{Allocated Bandwidth per slot} = \frac{1.536 \times 10^6}{24} = 64\,000 \text{ bps}$$

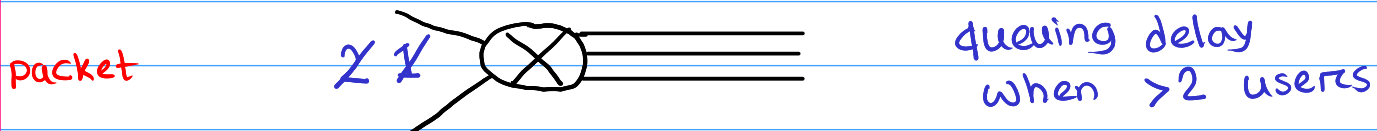
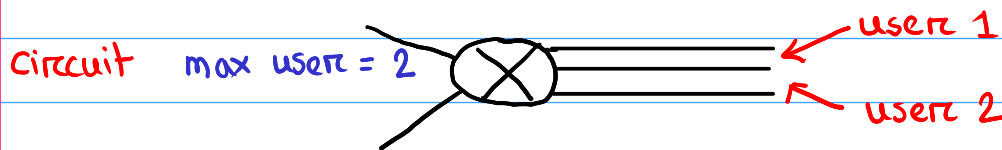
$$\text{File transmission time} = \frac{640\,000}{64\,000} = 10 \text{ s}$$

$$\text{Total time} = 10 + 0.5 = 10.5 \text{ s}$$

- ✓ **Exercise 5:** Suppose users share a 2Mbps (Megabits per second) link. Also suppose each user transmits continuously at 1Mbps when transmitting, but each user transmits only 20 percent of the time.

a) When **circuit switching** is used, how many users can be supported?

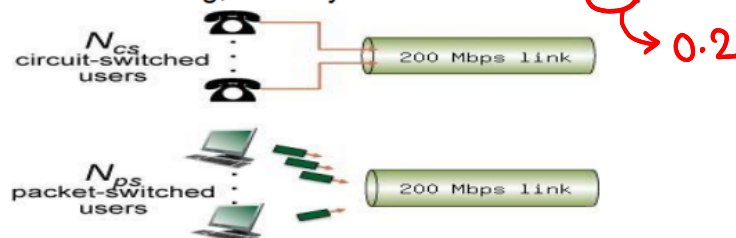
b) Suppose **packet switching** is used, why will there be essentially no queuing delay before the link if two or fewer users transmit at the same time?



$$\text{No of channels} = \frac{2 \text{ Mbps}}{1 \text{ Mbps}} = 2$$

- ✓ **Exercise 6:** This question requires a little bit of background in probability (but we'll try to help you though it in the solutions). Consider the **two scenarios** below:

- A circuit-switching scenario in which N_{cs} users, each requiring a bandwidth of 25 Mbps, must share a link of capacity 200 Mbps.
- A packet-switching scenario with N_{ps} users sharing a 200 Mbps link, where each user again requires 25 Mbps when transmitting, but only needs to transmit 20 percent of the time.

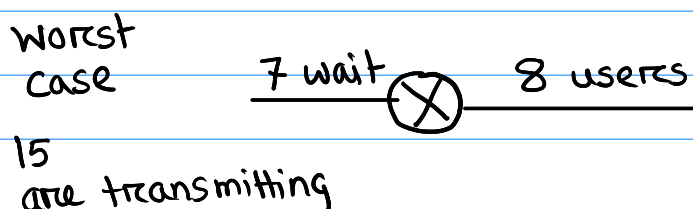


Answer the following questions:

- When **circuit switching** is used, what is the **maximum number of circuit-switched users** that can be supported? Explain your answer.
- For the remainder of this problem, suppose **packet switching** is used. Suppose there are **15 packet-switching users** (i.e., $N_{ps} = 15$). Can this many users be supported under circuit-switching? Explain.

a) $\text{No of channel / no of user} = \frac{200 \text{ Mbps}}{25 \text{ Mbps}} = 8$

b) $15 > 8 \therefore \text{No}$



- c. **What is the probability** that a given (specific) user is transmitting, and the remaining users are not transmitting?
- d. Find the probability that at any given time, **all 15 users** are transmitting simultaneously.
- e. **What is the probability** that more than 8 users are transmitting? Comment on what this implies about the number of users supportable under circuit switching and packet switching.

given/specific

☒	☐	☐	
0.2	0.2	0.2	transmit
0.8	0.8	0.8	not "

$$= 0.8 \times 0.2 \times 0.8$$

$$= \boxed{}$$

any

① ☒ ☐ ☐

$$= 0.2 \times 0.8 \times 0.8$$

② ☐ ☒ ☐

$$= 0.8 \times 0.2 \times 0.8$$

③ ☐ ☐ ☒

$$= 0.8 \times 0.8 \times 0.2$$

sum of all

$$= 0.384$$

Binomial Distribution

trial $\begin{cases} \rightarrow \text{success/transmit} \rightarrow p \\ \rightarrow \text{fail/not transmit} \rightarrow 1-p \end{cases}$

$$\boxed{{}^nC_r \cdot p^r \cdot (1-p)^{n-r}}$$

$\begin{cases} n \text{ trials} \\ r \text{ success} \end{cases}$

$$n = 3$$

$$r = 1$$

$${}^3C_1 \cdot (0.2)^1 \cdot (1-0.2)^{3-1}$$

$$= 0.384$$

specific

(c)

Total = 15
Transmit = 1
Not " = 14

$$0.2 \times (0.8)^{14}$$

(d)

All 15 users :

$$(0.2)^{15}$$

(e)

For 9 out of 15 :

$${}^{15}C_9 \times (0.2)^9 \times (0.8)^{15-9}$$

10

11

12

⋮

Generalised

$$\sum_{i=9}^{15} {}^{15}C_i (0.2)^i (1-0.2)^{15-i}$$

$\approx$ very small value